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First Snubbing Unit Perf, Wash and Cement Provided a Circulating Path from Anomaly to Surface to Ensure Long Term Isolation for a Strategic and Complex Wellsite

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Abstract

This document is an in-depth case study on the application of a snubbing unit for plug and abandonment (P&A) operations. The study highlights the successful execution of two sustained annulus pressure (SAP) remediation procedures within a single run, thereby eliminating the need for a conventional rig. The operation achieved complete rock-to-rock isolation and permanent well abandonment through the implementation of advanced techniques, including strategic perforation, washing, cement circulation, and precise cement plug placement.

The operation involved two remediation zones. The shallower zone was perforated to establish circulation to the surface, facilitating effective cement placement. Cement was then circulated to the surface and strategically displaced to form a continuous plug, ensuring full rock-to-rock isolation. This method aligns with the principles of the perforate, wash, and cement (PWC) technique, which has been recognized for its efficiency in establishing annular barriers without the need for section milling. PWC's effectiveness in curing SAP is further supported by its ability to restore annular barriers and improve well integrity, making it a cost-effective solution for P&A operations.

Snubbing units, they are very demanded for location with difficult and complex accessibility, in this case additionally to this limitation the well was located extremely near a very sensitive area, holding strategic constructions and equipment, options were analysed and with the location surface limitations, the very extensive location preparations required for a rig to mobilize, it was decided to look for alternatives. The first snubbing unit was mobilized to the country and with the limited capabilities to performing heavy well interventions under pressure, minimum pumping equipment to maintain well control throughout the operation a very robust intervention design and program was created. This approach eliminated the need for a traditional rig, reducing operational time and costs and more important allowing the required and committed intervention.

The successful implementation of this rig-less P&A operation demonstrates the viability of using snubbing units for remote, limited access, reduced real state and sensitive areas with complex well abandonment tasks. The approach not only ensures environmental compliance by achieving complete zonal

isolation but also offers significant cost and time savings. This case study contributes valuable insights into alternative P&A methodologies and techniques, emphasizing the potential of snubbing units and new technology in modern well abandonment operations.

Perforate, Wash, and Cement (PWC) is a proven technique for remediating SAP, also known as Sustained Casing Pressure (SCP), which is a common well-integrity issue in aging oil and gas wells. SAP occurs when pressure builds up in the annular space between casing strings due to compromised cement barriers, leading to potential environmental and safety concerns.

In summary, PWC is a reliable and efficient method for addressing SAP issues, offering significant time and cost savings while ensuring environmental compliance and long-term well integrity and it is possible to deploy this technology with a rig or even for rig-less operations.

The snubbing unit plus Perf, Wash and Cement method

The snubbing unit deploying a Perf, Wash and Cement technology offers an effective alternative to traditional rig-based section milling by creating a new annular seal without removing any section of the casing and without excessive metal debris creation and accumulation which are always very difficult and dangerous to handle. A snubbing unit is a specialized hydraulic system with multi-layered pressure control devices and components, used in the oil and gas industry to perform well intervention operations on live wells, it moves tubulars in and out of the well under pressure, (wells that are still under pressure) without the need to kill the well using heavy fluids that could therefore affect the rheology hierarchy during circulation, injection or cement operation.

A snubbing unit is a type of hydraulic workover unit (HWU) designed to insert or remove tubulars (pipes) into or from a wellbore while the well remains pressurized. This is achieved using a combination of hydraulic jacks, slips, and blowout preventers (BOPs) to control the pipe and maintain well control throughout operation. Unlike coiled tubing or wireline, snubbing units can handle heavier loads and are suitable for high-risk or heavy-duty tasks such as: Fishing operations, installing velocity strings or liners and/or well Plug and abandonment (P&A):

The use of the Snubbing units in well P&A allows us to have a hydraulic workover system with pipe movement in and out of a pressurized well by exhibiting sustained annulus pressure (SAP) mitigating in this way any risk of loss of well control during the Perf, Wash and cement operation, the process for these case studies involves three key steps:

1. **Perforation:** Perforations can be done via TCP or mechanically, in this novel paper a Mechanical system was used to avoid the use of explosives devices due to the strategic, highly regulated, restricted, complex location with limited access.
2. **Washing:** The annular space is cleaned to remove debris, old cement, and other obstructions, ensuring a clean surface for cement bonding. This method allows us to completely access the shallower section and establish circulation to surface, giving additional assurance of the isolation work performed on this well.
3. **Cementing:** Cement is pumped through the perforations to fill the annular space, creating a new barrier that restores well integrity. For this first-time snubbing unit project, it presented several technical challenges that were mitigated through engineering and alternative methods such as using the SU mud pump to carry out this service, special slurry designs to remediate the casing while taking into consideration the additional limitation of a SU and the potential additional times and operations.

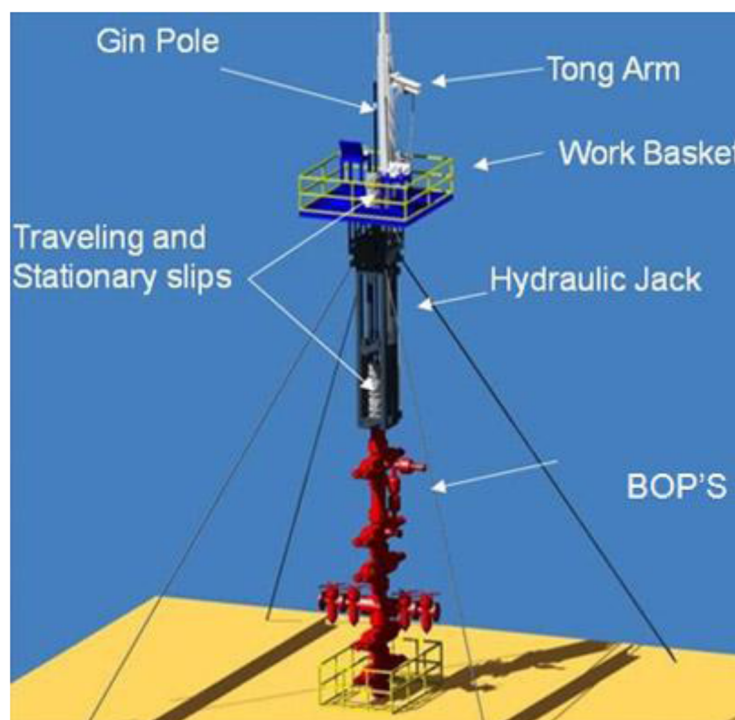


Figure 1—Illustrates a typical Snubbing Unit rig up

The Perf, Wash and Cement (PWC) technology is an effective alternative to the conventional approach of section milling or cut and pull. Thanks to implementation of PWC Technology there is a high potential of saved time, costs and reduced HSE hazards associated with Section Milling operations as well effective and long-lasting curing of Sustained Annulus Pressure.

Perforate, wash and Cement (PWC) Technique

The Perforate, Wash and cement (also known as PWC) is a technique for perforating a casing section, washing the annular space behind casing, removing cement, mud, barite, etc., then placing a new cement layer in the annular space and inside the tubing typically of 60m (200ft) as primary and secondary barrier. The effect is a rock to rock, cross sectional well barrier across the perforated section. If formation is behind pipe, then a rock-to-rock barrier is created. PWC is predominantly performed over a single casing section however there are dual casing scenarios where two annular spaces are PWC'd.

Traditionally, the creation of a rock-to-rock cement plug begins with milling a section of casing or liner. But removing hundreds of pounds of steel takes time and is often challenging; the resulting swarf requires special handling equipment and treatment. Also, squeezing cement into perforations without washing and purging the perforated zone is leaving a lot to chance. The PWC technique with close loop system (dual-inverted swap cups that create a pressurized chamber downhole to jet fluids and cement through perforations) eliminates the need to mill casing or liner, and it provides a clean and open annular space in which to place cement. Operating the PWC tool with close loop system is straightforward. It's one-trip operation consists of perforating the section, setting a hydraulic base for cement, hanging off the gun string; thoroughly washing the perforated section; then placing the cement plug. If required, drilling and logging the annular cement, and replacing the cement plug within the casing can also be performed. During cleaning, high-circulation rates enable improved efficiency and more effective removal of debris from the perforated zone and out of the well.

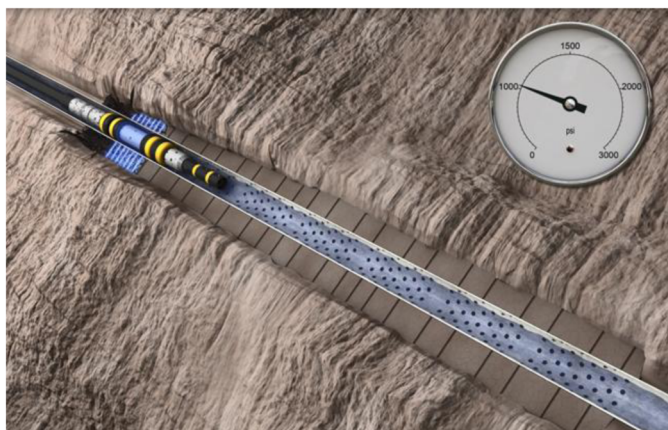


Figure 2—Illustrates a top-down washing pass at 1,000psi

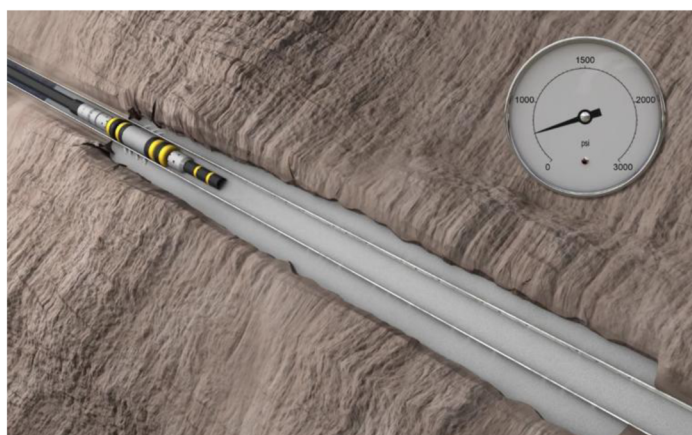


Figure 3—Illustrates a bottom-up cementing pass at 500psi

Over the last few years, the PWC technique has been used successfully for cement remediation and curing Sustained annular pressure (SAP) which is often caused by poor cement bonding or channelling behind casing, the PWC technique addresses this challenge by removing poor cement material in the annulus during the washing passes and by replacing it with fresh cement during the cementing pass, to finally establishing a cross sectional cement barrier using the closed loop system.

Perforate, wash and Cement Job Preparations in a Snubbing Unit

1. Before Running In Hole (RIH): It's important to assess the condition of the well and casing prior to RIH. Although not strictly required, performing a scraper or cleanout trip before running the tool is recommended (especially after cementing operations or perforations done with charges that leave large burrs on the internal wall of the casing) to help ensure a smooth and safe descent. Set shear values for solid and ball seats according to the expected pumping and operating pressures (typically, set a shear value of ± 3000 psi for 1.75" and 1.375" ball or solid seats).
2. Well Fluid Preparation: The well should be displaced to filtered brine before running the mechanical perforator to ensure optimal tool performance. High solids content can cause tool malfunction; if filtered brine or a filtration unit isn't available, and well control permits, fresh water may be used.
3. Selecting the Perforated Interval: Base your planned perforated interval on available log data when possible. If logs are available, optimize your selection to ensure the chosen section is suitable for both washing and cement placement. Pay close attention to formation patterns, stringers, clay content, and cement bond integrity.

4. Tally and String Verification: Verifying and correlating drill pipe (DP) tally with blank sections in the perforated area, including TCP connections and interval tops/bottoms. Ensure all crossovers meet API/Oilfield standards (avoid square internal edges). Minimum DP drift: 2". If Kelly valves or TIWs are used, ensure correct space-out for cementing. Confirm no floats are installed in the DP string.
5. Check for Restrictions: Identify restrictions such as liner laps and crossovers. Take special care while passing through these areas to protect the Bottom Hole Assembly BHA.
6. Ensuring Efficient Connections: Make all connections fast and safe during the washing sequence to prevent debris from settling.
7. Calculations: Perform all necessary calculations, including displacement for pump-and-pull cycles.
8. Equipment Verification: As soon as the crew arrives at the subbing unit, cross-check equipment and BHA measurements using checklist drawings, and inspect for damage. Confirm setting balls correspond with ball seat sizes.
9. Cement fundament: When selecting the option of using either a drillable or retrievable bridge plug and cement on top as the base for the subsequent PWC job, the cement base may be established before or after perforations are done with the mechanical perforator. Then the cement base plug must be properly spotted, tagged, and pressure tested prior to running the mechanical perforator in hole to punch/cut the PWC section. In all cases, the cement base should be positioned as close as possible to the bottom perforation, following standard guidelines: the cement plug should extend 30-50 feet below the bottom perforations, be dressed, and then tagged to confirm placement. If a drillable plug is used, it should be set 30 feet beneath the bottom perforation, while alternative cement-retaining devices such as a Cement Support Tool should also be installed 30 feet below the bottom perforation. For retrievable bridge plugs, the plug must be set at least 100 feet beneath the bottom perforation, with a high-visibility pill placed on top to confirm location. These procedures ensure a stable foundation for subsequent perforation and cementing operations.
10. Cementing Guidelines: For the PWC applications, it is normally recommended the G type cement, and the recommended densities are 16 -16,2ppg, since that is the most stable density, so that means that less chemicals are needed to make a good slurry. If for any special case lower weight of cement is needed, the overall recommendation is to have a cement with good stability, good rheological hierarchy for Mud vs Spacer Vs cement and acceptable gel values.

For optimal cement performance in PWC operations, a gas-tight slurry with low rheology and low gel strength should be employed, ensuring slow gel development during extended static intervals exceeding typical balanced plug durations. Prior to cement placement, a TT on-off-on test (this test simulates the intermittent pumping conditions that may occur during a real cementing job) with a pause of 30-60 minutes is recommended to verify continued pumpability in case of connection delays. Compatibility tests between the cement and standard spacers are essential; these should include assessments of cement rheology under spacer contamination, as well as TT and UCA testing with 20% spacer contamination. The results of UCA testing serve as guidance for establishing safe wait-on-cement (WOC) times prior to drill-out or tag operations. During the cementing operation itself it is advisable to use both Kelly Cocks and close the BOP on connections. To reduce contamination risk, placing at least 165 ft (50 m) of cement above the top shot perforation is recommended to ensure pumping and pulling above top shot after shearing out the ball seat.

11. Circulating clean after Pump & Pull Cement Job: When circulating clean after the cement job. The PWC System needs to be pulled in safe distance above theoretical top of cement (calculation based on nominal gauge hole). This is to make sure no cement is circulated to the surface, and that the cement that is displaced is left undisturbed to build lateral strength.
12. Heave compensator: On floating rigs it is essential to use heave compensator through any restrictions and during PWC Tool operation. Make sure that the compensator is in mid-stroke before use, to avoid the compensator to fully stroke.

13. **Mechanical Perforator Usage:** for the case studies in this paper, a mechanical perforator was used in the PWC operation, a mechanical perforator is a non-explosive perforation tool that replaces conventional TCP. Targeting 150 activations across a 150-ft interval optimizes annulus washing and debris removal.

Mechanical Perforator as part of the Perforate, wash and Cement

The case studies in this paper were designed and executed using a Mechanical Perforator, it is a pressure-activated, repeatable, and explosives-free tool designed to replace conventional TCP (Tubing Conveyed Perforating) guns in wellbore operations.

The mechanical perforator delivers a range of operational advantages ideally suited for modern well interventions. By eliminating explosives, it streamlines logistics and safety requirements, removing the need for transport permits and reducing overall risk. The tool's pressure-activated, repeatable mechanism allows for multiple perforations in a single run, achieving a rapid punch-to-punch cycle time of approximately 90 seconds at 0.5-meter spacing. Engineered for compatibility with common casing sizes such as 9 5/8", 10 3/4", and 13 3/8", the perforator is highly versatile, enabling seamless integration with plug and abandonment (P&A) and slot recovery operations, including deployment alongside the Perforate, wash and Cement (PWC) system. These features collectively enhance operational efficiency and flexibility in complex well abandonment scenarios.

The mechanical perforator is utilized across a range of well intervention applications, including punch, circulate, tag, and shear operations—particularly where production packers and oil-based mud (OBM) environments are present. It also serves in annulus verification when used alongside the formation testing and verification systems and supports casing cleaning and recovery through compatibility with casing cut and pull tools. Most notably, the mechanical perforator is integral to the Perf, Wash, and Cement (PWC) process, facilitating efficient and reliable abandonment solutions in complex well scenarios.

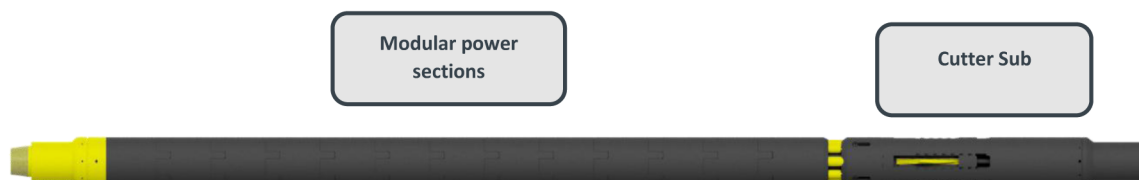


Figure 4—Illustrates a typical Mechanical Perforator Tool

For the case studies in this paper, the mechanical perforator was run in a two-trip system, it involved conducting all perforations on the first trip, followed by running the PWC tool and establishing the cement fundament in a second trip. During the first trip, the required interval is punched, while the second trip includes setting the cement base below the lowest perforation, thoroughly washing and cleaning the perforations with the PWC tool and subsequently placing the spacer and cement slurry across the perforated section. This approach eliminates the need for explosives permits related to import, storage, or use, streamlining logistics and enhancing safety. The system enables rapid deployment, with each perforation completed in just 90 seconds at 0.3-meter (1ft spacing for these jobs) and generates four slits at each cut depth for optimal annular access and cement placement.

When making up the BHA at surface, only the mechanical perforator is made up and RIH, so little time is spent on the rig floor in connection with the make-up, that compared to the TCP rig-up times, the mechanical perforator rig-up times are a substantial rig time saver. When at depth of the perforations, the mechanical perforator will be placed at the depth of the lowest perforation and the first punch (4 holes) will be performed. The subsequent sequence will typically be performed at a distance of 0.3m-0.4m with a time use of

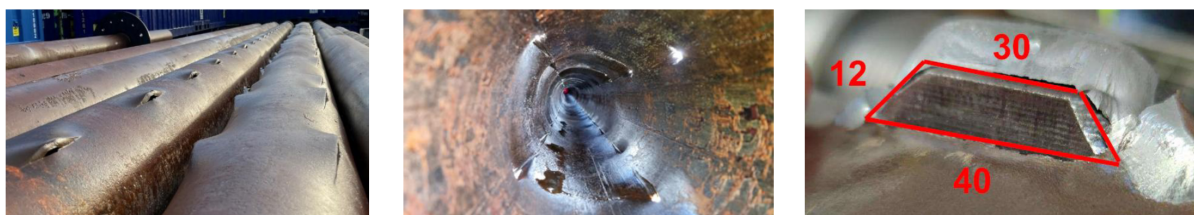


Figure 5—Illustrates a typical Mechanical Perforator Tool performance and cut dimensions in mm

Case Study: Operator A well B operations to establish well integrity

Well B was originally drilled and completed as a vertical dual water injector targeting the injection formations in the past. Instead of being put into service, the well was used as a reservoir observer during most of the well life. At the wellhead, pressure readings were recorded as zero psi (shut in) on both the long and short strings, while Annulus-A showed a sustained pressure of 400 psi and Annulus-B registered at 100 psi. Further investigation included bleed-off/build-up surveys, fluid sampling, wellhead seal tests, and noise/temperature logging were performed. Bleeding Annulus-A produced water with traces of gas, and the pressure would rebuild to 400 psi within 24 hours, indicating a steady flow at 4 psi. For Annulus-B, after an initial pressure bleed-down to 0 psi, the pressure did not return.

A leak detection study revealed a temperature anomaly at roughly 8,408 ft, suggesting a possible liner or casing leak, but no packer leak was detected. The plan was to permanently plug and abandon the well if no further objectives were identified by the Asset team B. Cement logging conducted during the operation showed potential favorable conditions for PWC application above 950 ft. In consultation with the operator team, the interval between 250 and 400 ft was selected for perforation, allowing the creation of an annular barrier above a water-producing formation at approximately 760 ft, which had been identified during drilling. For a successful PWC operations a cement base is required to be set 30-50ft below the perforations. Based on the ongoing operations, the current very safe, professional and responsible approach adopted by the operator pursuing the decommissioning of the asset for life, it was advised to the service company that the well was going to be abandoned with a solid rock to rock plug according to the logs and the cement plug program where cement plugs are set continuously from the deepest most section accessible in the wells to the required depth requested by the service company for the deployment of the PWC, so the use of the upper most existing cement plug would be the base for PWC operations instead of the typical available bridge plugs (TOC should be 30-50ft below bottom perforation), here in Figure 6, is the ultrasonic logging data of the interval between 250 and 400 ft to be PWC with indication of free pipe:

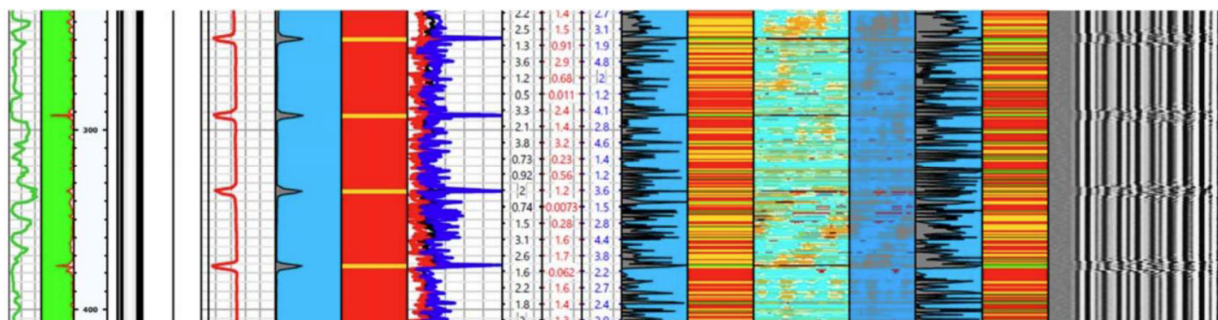


Figure 6—Ultrasonic logging data of the section to be PWC with indication of free pipe

With proper planning and optimized design and execution parameters, successful annular remediation can be achieved in large casing geometries using the Perforate, Wash and Cement Technique. This allows for more efficient and cost-effective well life extension and securing well integrity efficiently for operations or permanent decommission the asset as per codes of practice and as close to the natural original status, while

saving 65% on operating cost and at least 6 days on snubbing unit time. Other alternatives like cut and pull and section milling were not fully confirmed to be successfully possible to carry out with the snubbing unit.

As part of well preparation, verifications were done for any losses or inflow, to ensure they are controlled and minimized before initiating the SAP B remedial cement job using the PWC system. The well was stabilized prior to beginning the operation to support a successful outcome. A mechanical caliper (drifter) ran to determine the actual capacity of the cement string for accurate displacement and placement was also performed before starting the PWC operation. After setting the last cement base plug as a base for the SAP B PWC job, between 9-5/8in casing and 13-3/8in casing, tagging the cement base, marking the drill pipe, dressing cement, and performing a pressure test was also done.

The first step for the Well B PWC job was to perform the perforations using a mechanical perforator tool, a tool was run and the interval perforated from 400ft up to 250 ft. right after an injectivity test was performed with BOP closed and injecting fresh water at a rate of 1 bpm and monitoring the pressure in the casing and the annulus B. Observing and recording the pressures while verifying if there is circulation/connectivity between the perforations and the surface annulus B, checking for returns and pressures.

The second step for the Well B PWC job, after mechanical perforation of the well and verification of pressure stability, was the washing passes with the PWC tool, equipped with a closed-loop system of dual-inverted swab cups. The tool was picked up and ran in hole (RIH) to position the circulating ports 10 ft above the perforations at 240 ft measured depth (MD). A pressure test to 500 psi for 10 minutes was conducted to confirm tool integrity, ensuring that the cups were sealing properly and the solid ball seat was holding pressure in accordance with the operating procedures. Upon successful confirmation, the pressure was bled off to 200 psi and the tool was RIH below the top perforations at 250 ft. When pressure dropped to zero, the RIH operation was halted immediately, and the drill pipe was marked as a reference point for the top perforation. Subsequent washing passes across the perforations were executed as per the service company's protocols.

The third and final step for the WELL B PWC job was the cementing placement. The tool was run 5 ft below the bottom perforation at 405 ft, and pressure was applied up to 500 psi before slowly pulling into the perforations. When the pressure dropped to 0 psi, the pipe was marked as "BOTTOM PERF" and the depth was recorded as a reference point. The tool was then picked up to position the circulating port of the tool at 5 ft below the bottom perforations at 405 ft, and circulation was gradually established. Previously made drill string marks were used as references throughout the process, and circulation continued at least 1.5 bottoms up. The PWC tool was subsequently run in hole to 5 ft below the blank casing beneath the bottom perforation at 405 ft measured depth. An integrity test was performed at 500 psi for 5 minutes to confirm that the cups and solid seat/ball seat were holding pressure. Afterward, the tool was picked up to the bottom perforations at 400 ft, where circulation was again initiated gradually. Finally, the spacer then cement was pumped with the cement unit according to the prescribed pumping plan, as follows.

To complete the spacer displacement, a minimum volume of 39 bbl. of spacer was pumped while pulling ahead of the cement. Pump and pull operations were gradually initiated at the calculated speed and rate to ensure precise placement of the spacer behind the casing during pulling out across the 150 ft of perforations at 3.0 bpm and 17.85 ft/min. A total volume of 27 bbl., including excess per service company recommendations, was pumped and placed. The total spacer volume at this stage reached 40 bbl. The required tail spacer volume to displace the cement slurry was maintained at 5 bbl. plus an additional 1 bbl.

Batch mixing of 50 bbl. of cement slurry had been initiated in accordance with the cementing company, with all requisite materials and fluids staged in advance. Without initiating pumping, the PWC Tool was subsequently run in hole (RIH) and positioned at the bottom of the perforations (at 400 ft) maintaining a running speed of 2 min/joint to preserve consistent spacer placement. Pumping was deliberately withheld during this phase. The surface-side assembly, comprising the side-entry sub with TIW valve on top and LTV valve on the side, was rigged up. Surface lines were pressure tested to 3,500 psi against a closed LTV for a duration of 10 minutes. The cement unit's barrel counter was zeroed prior to the commencement of pumping.

A total of 50 bbl. of cement was pumped, with 10 bbl. displaced out of the tool prior to initiating pump and pull operations. Subsequently, 25 bbl. of cement was displaced from 400 ft to the top perforations at 250 ft, once in the last set of perforations the BOP was closed and the cement was circulated to the surface over the annulus observing returns in the gate valve, confirming a solid and complete isolation of the annulus from the perforations to the surface, with vigilant monitoring for returns. The operation adhered to a pre-established pump rate of 4.5 bpm and a pulling speed of 17.6 ft/min to ensure optimal placement. All cement was displaced, followed by a total of 3.2 bbl. being pumped out with mud, brine, or fresh water as the displacing medium.

The DP string was then pulled out of hole (POOH) to near the surface, and circulation was established to clean the tool. Volumes of spacer and cement returned to surface were meticulously recorded. Circulation was maintained for a minimum of three cycles, with continuous documentation of all returned spacer and cement volumes throughout the operation.

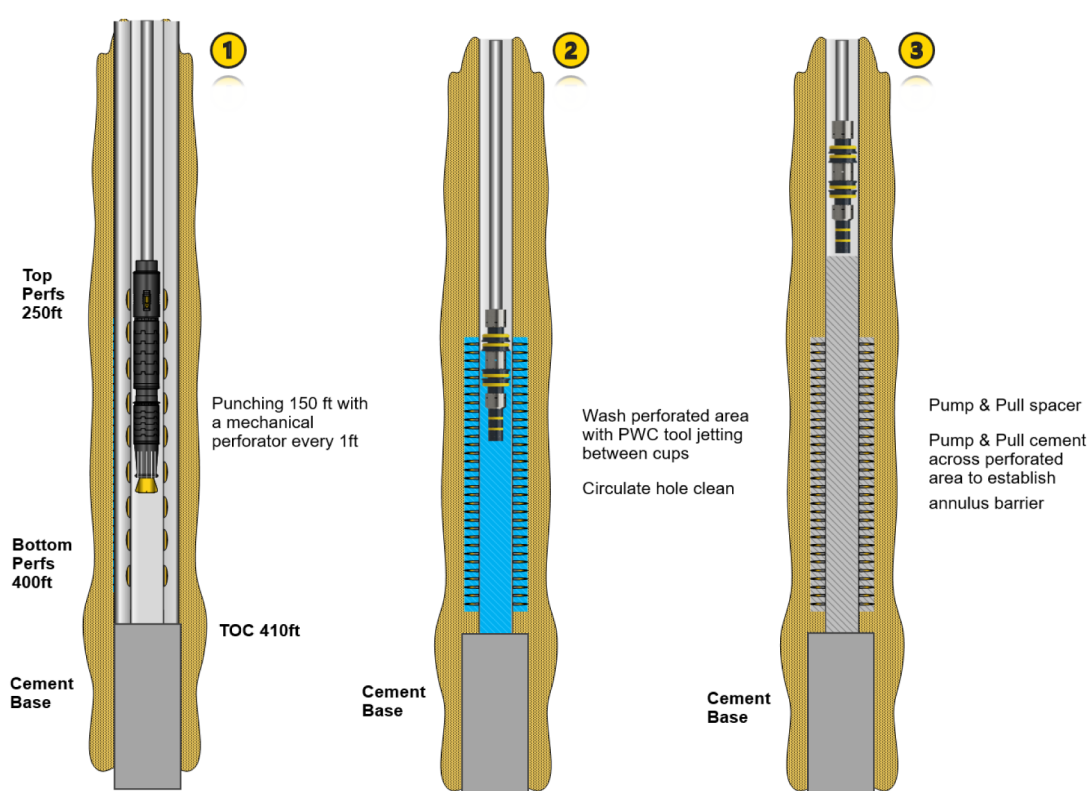


Figure 7—A 3 steps for the WELL B well Perf (mechanically), Wash and Cement job addressing sustained annulus pressure (SAP)

Conclusions

1. Professional and responsible permanent well decommission is a very important endeavour in the modern oil and gas industry to eliminate the potential for leaks, hazardous or polluting emissions aggravating the global warming. The PWC cement plug at Well B was tagged and pressure tested after a minimum of 24 hrs (WOC) without any disturbance from DHT. The results speak by itself, the SAP in annulus was cured and after monitoring for several weeks there was no buildup of pressure, the onshore operator team, recognized the PWC technology as a cost-effective technology to remove SAP in this well. After this

Some of the most relevant features for the Perf, Wash and Cement (PWC) system are:

2. Perforating guns can be installed below the PWC Tool if sufficient rathole is available and application is well P&A
3. Perforations can be performed using conventional TCP guns or mechanical perforators
4. Wellbore perforating / cleaning / cementing can be done in one run
5. Simple modular design makes it easy to change configuration on the rig site
6. Ball catcher incorporated in the System leaves no activation balls in hole
7. Built in bypass system, no surge or swabbing while RIH and POOH
8. High flow capacity ensuring wellbore clean up
9. Dependi
10. ng on the application, the PWC system can be optimized to further reduce operational costs and risk exposure.

References

1. N. Hurtado; F.Z. Bermudez; A. Alzaabi; M. BinKalban; M. Karmstaji; K. Roicki, Extending Well Life and Securing Well Integrity Efficiently SPE-222408-MS, ADIPEC, Abu Dhabi, UAE, November 2024.
2. B. T. Torvestad, A. Fliss, F. Z. Bermudez, T. A. Stokkeland, T. Fossdal, M. C. Carlsen, Perf Wash and Cementing Technique Remediate Cemented Annulus to Achieve High Quality Bond Across the Entire Interval in Complex Slot Recovery Operation SPE-210783-MS, APAC SPE conference Adelaide October 2022
3. T. A. Stokkeland, E. Malde, P. O. Staveland, Y. Frayland, S. Johnsen Dybvik, J. Jacobsen, T. Kibsgaard Vatn, S. Opedal (2020), Innovative One Trip System Helps Qualifying Creeping Shale as Permanent Barrier for Plug and Abandonment of Wells on the Gyda Field, SPE-200755-MS, SPE Norway Subsurface Conference, Virtual, November 2020.
4. A. Lucas, T.A. Stokkeland, D. Jackson, S. Ingram (2019), Improving the understanding, application and reliability of the Perforate, Wash and Cement technique through the use of cement bond logs, tool enhancements and barrier verification via annular pressure monitoring, SPE-191528-MS SPE Annual Technical Conference and Exhibition in Dallas, Texas, USA, 24-26 September 2018.
5. A. Lucas, T.A. Stokkeland, D. Jackson, S. Ingram (2019), Improving the understanding, application and reliability of the Perforate, Wash and Cement technique through the use of cement bond logs, tool enhancements and barrier verification via annular pressure monitoring, SPE-191528-MS SPE Annual Technical Conference and Exhibition in Dallas, Texas, USA, 24-26 September 2018.
6. Norwegian Oil and Gas Association | The Federation of Norwegian Industries. 2013. *NORSOK Standard D-010: Well integrity in drilling and well operations, Revision 4*
7. Barclay, I. et al: "The Beginning of the End: A Review of Abandonment and Decommissioning Practices," *Oilfield Review* (Winter 2001), **13** (4), 28–41